

# Georgia Institute of Technology

## CS 4365/6365: Introduction to Enterprise Computing

Summer 2026

### Project Checkpoint 1 Report

**Group:** 4

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**Project Name:** NORP Food Assistance Need-Capacity Gap Explorer

### Context and Related Work:

Prior NORP work focused on NL2SQL/RAG chatbots so non-technical researchers could query nonprofit data. Across three semesters this consistently drifted into SQL-generation benchmarking rather than the actual goal of surfacing sociological correlations. This semester's course direction is an agentic data exploration layer that cleans data, verifies its own work, and computes correlations to surface trends.

Our project applies this to a concrete question: where does food-related community need outpace nonprofit capacity to address it? We join nonprofit financial/category data with county-level need indicators to compute a simple, explainable gap score, and use an LLM-guided layer to propose and rank candidate correlations rather than just to translate questions into queries.

### Project Plan (Plan)

**Research question:** Can an agentic workflow identify U.S. counties where food-related need is high but nonprofit food/human-service capacity is relatively low?

**Secondary question:** Which indicators correlate most strongly with nonprofit capacity across counties? (Unit of analysis: county, joined on FIPS code)

### Where the Agent Acts

The agent does not work off a static script. Instead, there are three points where the system makes autonomous decisions:

1. Data-quality triage: the profiler classifies each table/join as usable, usable\_with\_warning, or drop, and the pipeline acts on that classification without manual intervention (for example, auto-excluding a state from a join if its match rate falls below a set threshold, then logging why).

2. Correlation candidate generation: rather than a hardcoded list, an LLM step proposes candidate need-vs-capacity variable pairs from the available schema, which Python then tests and ranks (this is the “exploratory, not retrieval” behavior)
3. Self-verification gate: before any plot or summary is produced, a rule-based + LLM check reviews the join and data-quality reports and issues a `proceed` / `proceed_with_warning` / `stop` verdict, which determines whether downstream steps run.

## Data Sources (verified via NORP Metabase / NCCS database)

We confirmed real tables and join paths directly in Metabase rather than assuming availability:

1. Nonprofit capacity: `NGOs_with_categories` (EIN, county, NTEE code/category — 3.42M orgs, ~320K in food/human-service NTEE codes:  $K^* = 40,080$ ,  $P^* = 280,076$ ) joined via EIN to `F9_P01_T00_SUMMARY_2022` (131,587 2022 990/990-EZ filings — revenue, expenses, net assets, employees).
2. Community need: `disadvantaged_communities` (72,742 census tracts, clean `county_fips` key, includes `food_desert_pct`, poverty, housing burden) aggregated to county; supplemented by `Poverty_Rates_2023` and `nccs_crosswalk_economic` (ACS poverty/unemployment/income) as needed.
3. Fallbacks only if needed: IRS EO BMF (nonprofit registry) and direct USDA Food Environment Atlas/ACS pulls. We demoted this from primary since the Metabase already covers this ground.

## Known Data Risk: County Join

The capacity side keys on county names while the need side keys on FIPS codes. We tested this join directly: `county_fips_lookup` is missing Florida entirely, and Connecticut nonprofits use planning-region names instead of counties. Together, these account for the large majority of join failures we found. **Plan for fix:** patch the crosswalk with FL FIPS codes and a manual CT planning-region mapping (8 regions) before Checkpoint 2. Then, any remaining unmatched counties will be logged/excluded rather than hand fixed.

## Need-Capacity Gap Score

**Need Score** = average z-score of available need indicators (poverty rate, food desert pct, housing burden, etc.). **Capacity Score** = average z-score of available capacity indicators (nonprofits per 10k residents, net assets per capita, revenue per capita). **Gap Score** = Need Score – Capacity Score. This is a triage signal, not a causal claim. Essentially, a high gap means worth investigating further, not proof of nonprofit failure.

## Non-Goals

1. No chatbot interface; no Spider/WikiSQL-style NL2SQL benchmarking.
2. No causal claims about nonprofit effectiveness (no spatial/multilevel modeling at this stage).

3. No forcing of unreliable joins, we will log and drop.
4. Python outputs are the source of truth for all statistics. The LLM only proposes candidates and writes summaries.

## Project Deliverables

| Deliverable                   | Description                                                                                                                                                                                  | Technical Stack             |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Profiler & Quality Gate       | Loads nonprofit and need-side data, classifies each table/join as usable, usable_with_warning, or drop, and decides proceed / proceed_with_warning / stop before downstream steps run.       | Python, pandas              |
| County Capacity & Need Tables | Aggregates NGOs_with_categories and F9 990 filing data, and tract-level need data (disadvantaged_communities), to county level.                                                              | Python, pandas              |
| Joined Gap-Score Panel        | County FIPS join of capacity and need tables; computes need_score, capacity_score, and gap_score.                                                                                            | Python, scipy               |
| Correlation Agent & Reporting | LLM proposes candidate need-vs-capacity variable pairs from the schema; Python tests and ranks them, generates top-gap county plots, and writes a findings summary from the computed tables. | Python, matplotlib, LLM API |

## Project Milestones

| Checkpoint | Milestone | Technical Scope & Deliverables | Work Splitup | Status |
|------------|-----------|--------------------------------|--------------|--------|
|            |           |                                |              |        |

|                     |                                                                                          |                                                                                                      |      |             |
|---------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------|-------------|
| <b>Checkpoint 1</b> | Scope project, verify Metabase data sources, test county join feasibility, repo skeleton | Repo structure, Metabase verification, GitHub setup                                                  | Both | In progress |
| <b>Checkpoint 2</b> | Build profiler, capacity/need tables, fix FL/CT crosswalk, produce joined county panel   | Ishaan: profiler, join logic. Gowtam: capacity/need table builders.                                  | Both | Planned     |
| <b>Checkpoint 3</b> | Gap scores, correlation discovery agent, plots, findings summary                         | Ishaan: pipeline automation, output organization. Gowtam: scoring, correlation agent, summary draft. | Both | Planned     |

## Current Progress Report (Match)

- Reviewed the NORP pivot and the reproducibility exercise, then scoped a concrete project (food need vs. nonprofit capacity) instead of a generic correlation workflow.
- Logged into the NORP Metabase and verified, not assumed, our data plan. We confirmed real tables, columns, and row counts for both nonprofit (NGOs\_with\_categories, F9 990 tables) and need-side data (disadvantaged\_communities), matching the Checkpoint 1 milestone of finalizing the dataset plan.
- Next two weeks: patch the FL/CT county crosswalk gap, and implement the profiler and capacity/need table builders so Checkpoint 2 has a real joined panel rather than a plan.
- Change from original plan: stress-testing the county join surfaced a concrete, fixable gap (Florida missing from the county\_fips\_lookup crosswalk; Connecticut nonprofits use planning-region names instead of counties) that was not anticipated when the data sources were first chosen, and is now a tracked fix item for Checkpoint 2.

## Supporting Evidence (Factual)

- Team GitHub Repository: <https://github.com/ishaanbhardwaj2003/NORP-Food-Assistance-Need-Capacity-Gap-Explorer>
- Using the Metabase(NCCS database) we directly verified the following:
  - **NGOs\_with\_categories:** Contains 3,420,024 rows with unique EINs (0 duplicates confirmed) and a confirmed NTEE category breakdown (K\* food/ag/nutrition = 40,080 orgs; P\* human services = 280,076 orgs).
  - **F9\_P01\_T00\_SUMMARY\_2022:** Contains 131,587 filings for 2022 with revenue, expense, and asset fields. We confirmed that 990-EZ filers have null total assets but populated net assets, which informed our capacity metric choice.
  - **disadvantaged\_communities:** Contains 72,742 census tracts with a clean county\_fips key and food\_desert\_pct present for 99.6% of rows. This was independently re-verified by downloading and inspecting the full export in pandas.
- County join test: ran the proposed county-name-to-FIPS join directly in Metabase; identified Florida's absence from county\_fips\_lookup and Connecticut's planning-region naming as the dominant failure causes, with a documented fix plan for Checkpoint 2. Screenshots of sample rows from NGOs\_with\_categories and disadvantaged\_communities are included in the repository.

## Skill Learning Report

Data integration & profiling: inspecting real schemas (row counts, nulls, join keys) before writing pipeline code, and treating profiling output as a gate rather than a formality — we put this into practice by stress-testing the county join in Metabase instead of assuming it would work.

Agentic workflow design: separating where Python is authoritative (statistics, joins) from where an LLM adds value (candidate generation, summarization) instead of routing everything through one model call.

County-level data joining: handling FIPS/text-name mismatches and documenting, rather than silently absorbing, geographic join failures.

## Self-Evaluation

- Plan: 100%
- Match: 70%
- Factual: 70%

## LLM Feedback:

- Project Plan (Plan): Concrete and scoped, with real verified data sources rather than assumed ones. Improvement over a generic correlation-workflow plan. The “Where the Agent Acts” section makes the agentic design explicit rather than implied.
- Progress Report (Match): Strong for Checkpoint 1 because the team verified the data plan against the live Metabase instance instead of taking documentation at face value, including stress-testing the riskiest join. No pipeline code yet, which is expected at this stage.
- Supporting Evidence (Factual): The verified schema/row-count findings are real, independently checked evidence. The repository link is now in place; the team should keep committing starter scripts and sample screenshots so Supporting Evidence is fully self-contained by submission.

## Actionable Suggestions:

- Patch the FL/CT crosswalk gap before Checkpoint 2 so the join isn’t a surprise mid-pipeline.
- Commit starter scripts (load\_data.py, profile\_data.py) and README to the repo now to strengthen Supporting Evidence ahead of submission.
- Keep the LLM scoped to candidate generation and summarization — don’t let it compute statistics.